BTS–VII(SS)–05.17–0858 Reg. No.

B

B.Tech. Degree VII Semester Special Supplementary Examination May 2017

EB/EC 1705 (E4) MECHATRONICS
(*2012 Scheme*)

Time: 3 Hours Maximum Marks: 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) What is Mechatronics? Discuss on the scope of Mechatronics.
(b) Describe the input output relationship for rotational translation system.
(c) What is the need for bearings? Explain the different types of ball bearings.
(d) What is the need for servo motor? Explain the working of AC servo motor.
(e) Explain the basic structure of a robotic system.
(f) What are the classifications of NC systems?
(g) Discuss the importance of MEMS in mechatronics.
(h) What is derivative control? Explain how PD control is implemented using Op-Amp circuit.

PART B

(4 × 15 = 60)

II. (a) Define the following performance measures for a second order system. (10)
(i) Rise time.
(ii) Peak time.
(iii) Overshoot.
(iv) Settling time.
(b) Explain the input output relation of the rack and pinion. (5)

OR

III. (a) Obtain the relationship for the basic building blocks of mechanical and electrical systems. (10)
(b) Write short notes on dynamic response of systems. (5)

IV. Explain the following. (15)
(i) Rotary actuators.
(ii) Ratchet and Pawl.
(iii) Operation of CAM.

OR

V. Explain the following. (15)
(i) Stepper Motor.
(ii) Encoders.
(iii) Pressure control valves.

VI.	(a)	With a neat block diagram explain a CNC system.	(10)
	(b)	List the design consideration of NC machine tool.	(5)

OR

VII.	(a)	Explain the classification of robots.	(8)
	(b)	Explain the classification of NC system.	(7)
VIII.	(a)	With a neat diagram explain the structure of PLC and mention the selection criteria for the same.	(10)
	(b)	List the advantages and applications of MEMS.	(5)

OR

IX.		Explain the various closed loop control modes in detail.	(15)

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